Peiyuan Zhang

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Research Interests

Macroeconomics, Computational Economics, Artificial Intelligence, Numerical Methods

Education

School of Economics, Peking University, Ph.D. Economics	2022 – Present
School of Economics, Peking University, B.S. Economics	2018 – 2022

Working Papers

Informed Actor-Critic Method for optimal control in Economic Dynamics.
with Bo Li and Serguei Maliar

This paper applies deep reinforcement learning to high-dimensional optimal control in dynamic economic models. We introduce an informed actor-critic framework that embeds structural equations and constraints directly into the policy optimization loop. Unlike standard model-free reinforcement learning algorithms that treat the economy as a black box, the iAC method leverages analytical economic transitions. Applications to benchmark consumption-savings, Krusell-Smith, and overlapping-generations economies demonstrate that iAC eliminates unnecessary exploration noise and provides a precise, scalable, and sample-efficient framework for complex macroeconomic control problems.

Solving Discrete-Continuous Dynamic Choice Models: A Soft-to-Hard AI Solver with an Application to Sovereign Default.
with Bo Li and Serguei Maliar

This paper introduces a soft-to-hard (StH) deep reinforcement learning framework to solve discrete-continuous dynamic choice models. To overcome vanishing gradients caused by non-differentiable kinks, we augment an actor-critic architecture with distinct training and evaluation modes. During training, hard discrete switches are replaced with soft choice probabilities, allowing gradients to pass through value branches and policy networks. The smoothing is temporary: the temperature parameter is annealed toward zero, and evaluation returns to the original hard-switch environment. Applications to one- and multi-period sovereign default problems show that the StH solver delivers accurate, scalable solutions without relying on permanent taste shocks or random maturity assumptions.

Learning to Expect: Conditional Expectation Functions for Neural Economic Solvers

Neural and model-free methods provide flexible solution frameworks for dynamic economic models, but their Bellman targets still rely on conditional expectations of continuation values, commonly approximated by one-shot Monte Carlo draws that are then discarded. This project develops W-exp, a learned conditional expectation module that converts simulated shock information into a reusable expectation function over post-decision states and improves the expectation-construction step shared by many neural and simulator-based solvers.

AI Taxing AI: A Multi-Agent Reinforcement Learning Approach to Optimal Taxation
with Bo Li

Artificial intelligence (AI) is improving productivity, yet its economy-wide diffusion may reshape the labor market and the distribution of income. To address these issues, we develop a Multi-Agent Reinforcement Learning (MARL) model featuring endogenous skill accumulation. We find that while AI boosts aggregate output via rapid process of skill sharing, it displaces middle-skilled humans and increases inequality. However, optimal taxation can redistribute the technological surplus to achieve Pareto improvements. To find the optimal tax schedule, we employ Dual-Loop Multi-Agent Reinforcement Learning to account for both government's fiscal policy and agent behaviors. Furthermore, we demonstrate that a "Digital Commons" regime—combining universal access with real-time knowledge sharing—maximizes social welfare by mitigating the trade-off between efficiency and equity.

A Quantitative Assessment of Pension Reform in China: Contribution, Benefit and Delayed Retirement
with Bo Li and Yi Lu

How does delayed retirement policy affect agent behavior and the aggregate economy in China? To evaluate this policy, we develop a heterogeneous OLG model with endogenous retirement choice and the contribution/benefit rules of the social security system in China. We find that an increase in the lower contribution bound significantly boosts consumption by 2.34% and welfare by 1.06%, while simultaneously improving pension finances and income equity. In contrast, adjusting the upper bound has a negligible impact. Furthermore, introducing a delayed retirement policy yields additional gains, as it increases welfare and reduces the pension deficit. However, the policy's effects are heterogeneous across different groups. Finally, we examine an extension highlighting the importance of life-cycle considerations in policy design.

Forecasting Crude Oil Spot Prices Using a Transformer-BiLSTM Architecture with NRBO-based Hyperparameter Optimization
with Wenhui Huang et al.
R&R at North American Journal of Economics and Finance

A future vision of global crude oil markets is crucial for market participants and policymakers. However, making an accurate prediction of crude oil prices is always challenging, due to the difficulty in extracting both global and local information from historical price data and optimizing hyperparameter settings of the forecasting framework. In this research, a Transformer-BiLSTM-NRBO forecasting model architecture is proposed to overcome these challenges. First, a Transformer-based feature extraction module is adopted to enhance the capture of global information in the forecasting framework. Then, a BiLSTM-based feature extraction module is used to enhance the capture of local information. Finally, an NRBO algorithm is adopted to optimize the key hyperparameter settings of the previous modules. Experiments conducted on daily Brent and WTI crude oil spot prices from 2013 to 2024 illustrate the robust and outstanding forecasting capability of the proposed method. Research limitations and future directions are given.

Heterogeneity Models Introduce Time-varying Risk Aversion Affected by Economic Shocks
with Bo Li

We introduce a time-varying risk aversion to the Aiyagari model and link the time-varying risk aversion to the heterogenous shocks. We also introduce different types of individuals who change their risk aversion differently by shocks. The new model enriches the variety of individual heterogeneity without changing the random shocks and fits better gini coefficients for wealth. And we find that the new model obtains the above effect through two mechanisms: 1) more parameter dimensions 2) influence the saving behavior.

Work in Progress

Reduced-State Deep Learning for Heterogeneous Agent Models

This project develops a reduced-state neural solution method for heterogeneous agent models with aggregate uncertainty. Instead of feeding the full cross-sectional distribution into policy and value networks, we approximate the aggregate state using a low-dimensional vector of moments, such as mean capital and its dispersion, that is updated via simulation. The approach bridges the gap between belief-based approximations and full-distribution neural methods while maintaining computational tractability for high-dimensional HAM environments.

Publication

Asset Pricing in China's Stock Market (with Shuaiyu Jiang)
Finance & Economic Vision, 2022

This paper uses traditional machine learning methods and deep neural networks based on both firm-specific characteristics and macroeconomic variables to price China's A-share stock market. The main contributions are as follows: We give the stochastic discount factor a flexible form and compare different models' performances. Since the Chinese government adopts various policies to maintain financial stability, we borrow the idea from generative adversarial network to find the true SDF by selecting moment condition that minimizes return volatility. Additionally, we compare this model's performance with Chen's work and find that this model can obtain higher Sharpe ratio and R Square.

Project Experience

NSFC, Original Exploration Program, "Construction of Macroeconomic Models Based on Intelligent Algorithms and Agents and Their Application in Policy Evaluation" — Core Participant

NSFC, Young Scientists Fund Program, “Design of Optimal Property Tax Reform: A Quantitative Study Based on Heterogeneous Household Models” — Core Participant

Peking University Open Project of the National Intelligent Social Governance Experimental Base in Donghu High-Tech Zone, “Research on Macroeconomic Theories, Algorithms, and Models Based on AI — Scenario Simulation in Donghu High-Tech Zone” — Core Participant

New Engineering Interdisciplinary Youth Project, “Emergence and Evolution of Currency Based on Multi-Agent Learning” — Core Participant

Teaching Experience

Series of Literature Interpretation, Derivation and Replication on Solving HAM with Deep Learning

Organized reading group covering key papers; provided replication codes and derivations.

Principles of Macroeconomics (undergrad), TA Spring 2022 – 2025

Artificial Intelligence and Society (undergrad), TA Spring 2022

Computational Economics (grad), TA Spring 2022

Skills

Programming: Fortran, Python, Matlab, C/C++, CUDA

Languages: English (Fluent), Chinese (Native)

Interests: Photography

References

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